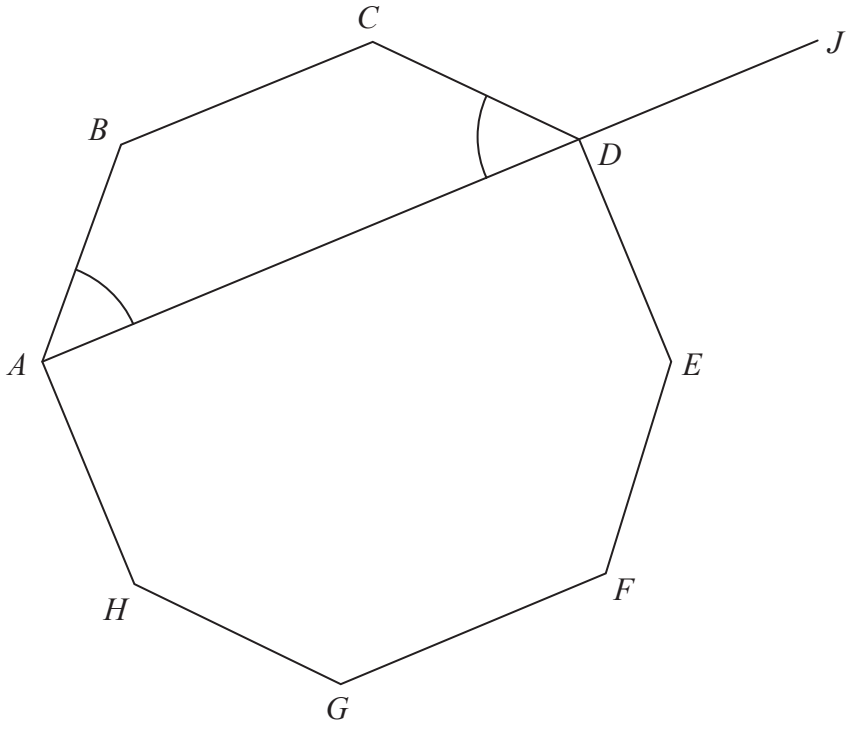
6



ABCDEFGH is a regular octagon.
ADJ is a straight line.

angle *BAD* = angle *CDA*

Show that angle *CDJ* = 135°

Interior angle = (8 – 2) × 180° ÷ 8.

1 mark
Regular-polygon method

= 135°, so ABC = BCD = 135°.

1 mark
Interior135°

ABCD is a quadrilateral: angle sum360°.

BAD + CDA = 360° – 135° – 135° = 90°.

1 mark
Find CDA=45°

Given BAD = CDA, so CDA = 45°.

ADJ is straight: CDJ = 180° – 45° = 135°.

1 mark
Straight-line conclusion

(Total for Question 6 is 4 marks)